

Backup Infrastructure Review, Remediation & Verification Report

Prepared for review with T-Net · covering 2026-07-07 → 2026-07-20

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Status: Service restored; verification complete; guidance requested

Executive summary

Between **2026-07-07 and 2026-07-15**, all Veeam backup jobs at KOR Structural failed silently following the expiry of the vCenter TLS certificate. The condition was identified on July 15 during a manual console review. A subsequent recovery attempt could not complete because of the interaction between the existing job design and available repository capacity.

All three protection tiers are now operating. KOR has restored local, off-site and file-server backups, patched and verified the backup server, configured failure notifications, and — for the first time in this environment — performed and documented restore testing.

Backup gap: 2026-07-07 → 2026-07-20 for KOR-FS01 (13 days); 2026-07-07 → 2026-07-20 for all other VMs (restored July 15–20)

Current state: Local nightly ✓ · KOR-FS01 dedicated chain ✓ · Off-site copy ✓ · Alerting ✓

Restore testing: 2 of 3 scenarios pass; the third has a proven, understood cause

Principal open items: Veeam version upgrade · mount server for deduplicated volumes · off-site re-seed for KOR-FS01 · repository capacity strategy

Purpose of this document: to share the complete technical picture and the evidence gathered, and to obtain T-Net's guidance on the four open items above so that a durable design can be agreed.

1. Incident timeline

Date	Event
Jul 7, ~17:00	Last successful backup run across all jobs (confirmed by restore-point files on disk).
Jul 7 / 8	vCenter TLS machine certificate expires. Veeam pins the trusted certificate thumbprint, so every nightly job and the 4-hourly infrastructure discovery begins failing. Backup coverage stops.
Jul 13	Windows Data Deduplication enabled in-guest on KOR-FS01 (E: volume). The initial optimisation rewrites approximately 10 TB of blocks and clears the volume's existing VSS shadow copies.
Jul 15, 11:09	vCenter issues a replacement certificate automatically (valid to 2028-07-14).
Jul 15, ~12:40	KOR identifies the outage during manual console review, re-accepts the new certificate, and restarts the job chain. The 8-day gap ends.

Jul 15 – 18	The restarted job processes 5 of 6 VMs successfully. KOR-FS01 cannot complete: in reverse-incremental mode the rollback file grows to 8.9 TB against 1.8 TB of remaining repository capacity.
Jul 18 – 19	Run stopped to avoid exhausting the repository. Session termination follows Veeam KB1727. vCenter subsequently confirms the temporary snapshot on KOR-FS01 was removed and that consolidation was not required.
Jul 19 – 20	KOR-FS01 protected via a new forward-incremental job to the Synology repository. Remaining jobs returned to schedule. Notifications configured. Restore testing performed.

2. Verified environment

All figures below were verified directly from the Veeam console, the Veeam configuration database, the repository filesystems, and vSphere between July 15 and July 20.

Backup infrastructure

Component	Detail
Backup server	KOR-BK01 — virtual machine, Windows Server 2019, 16 GB RAM, workgroup, PostgreSQL configuration database
Veeam version	12.0.0.1420 (v12 GA, 2023). Current shipping release is 13.0.2.29 (May 2026).
Licence	Enterprise Plus, rental via T-Net
Remote management	Veeam Availability Console agent, NinjaRMM, ScreenConnect / TeamViewer

Repositories

Repository	Path	Capacity	Free	Notes
Default Backup Repository	E:\Backup (KOR-BK01)	25.4 TB	1.82 TB	~23 TB is the KOR-FS01 chain (see §5); flags low-space warning every run
Synology105	F:\Backups (KOR-BK01)	26.4 TB	5.74 TB	Now hosts the new KOR-FS01 chain
Kor-Replication	Cloud Connect — remote.t-net.ca, path \kor	25 TB	11.47 TB	Off-site copy hosted by T-Net

Virtual infrastructure

Component	Detail
vCenter	192.168.1.9 — new certificate valid to 2028-07-14
Hosts / cluster	ESXi 192.168.1.10 and 192.168.1.16, cluster "Van-Cluster"

Primary datastore	UC3200 — 50.78 TB, 7.5 TB free
KOR-FS01	21.1 TB file server, Windows Server 2025, Windows Data Deduplication enabled on E: since 2026-07-13 (≈50% savings)

Jobs (current)

Job	Type / mode	Target	Objects	Status
Kor-FS01 NEW	Backup — forward incremental	Synology105	1 (KOR-FS01)	Active, nightly
Kor-VMs-New	Backup — reverse incremental	Default (E:)	3 (APP01, DC01, RDS01)	Active, nightly
Vcenter	Backup — reverse incremental	Default (E:)	1 (vCenter appliance)	Active, chained
Kor-Replication	Backup copy	Cloud Connect (T-Net)	2 source jobs	Active, chained
ManualNew	Backup	Synology105	4	Enabled but unscheduled; never run — origin / purpose to be confirmed

3. Remediation completed by KOR

#	Action	Date	Result
1	Re-accepted the new vCenter certificate; restarted the job chain	Jul 15	Infrastructure discovery and jobs reconnected
2	Stopped the non-viable reverse-incremental run per Veeam KB1727; verified snapshot removal	Jul 18–19	vCenter confirmed snapshot deleted; consolidation not required; no stray hot-added disks
3	Reclaimed system-drive capacity on KOR-BK01; applied Windows updates (4 security KBs); rebooted; verified services and rescanned repositories	Jul 19–20	C: from 20 GB to ~50 GB free; updates applied and server rebooted (after backups, none interrupted); all Veeam services healthy; repositories online
4	Created a dedicated KOR-FS01 job — forward incremental — targeting Synology105	Jul 19–20	5.79 TB written · ~126 MB/s · ~14 hours · 0 errors
5	Removed KOR-FS01 and the decommissioned BMZ-HO-APP003 / APP004 from Kor-VMs-New; re-enabled Kor-VMs-New, Vcenter and Kor-Replication	Jul 20	Full chain including off-site copy completed successfully
6	Configured Veeam email notifications to KOR	Jul 20	Verified delivering — sample job-completion alerts received July 21

Performance comparison — the same workload, two designs.

- **Reverse incremental** → **E**: ~29 MB/s, bottleneck reported as *Target*, ~23 TB of repository consumed, run did not complete.
- **Forward incremental** → **Synology105**: ~126 MB/s, 5.79 TB written, completed in ~14 hours with 0 errors.

Approximately **4× the throughput in roughly one third of the space**. This measurement informs question 1 in §6.

4. Verification — restore testing

No restore had previously been performed or documented in this environment. Three scenarios were tested on 2026-07-20. All restores were directed to a new location; no original data was modified.

Scenario	Result	Finding
1. KOR-FS01 file-level restore from the local pre-deduplication chain (E:, July 1–7 restore point)	PASS	The E: chain is genuinely restorable — its history is verified, not merely catalogued.
2. KOR-FS01 file-level restore from the off-site Cloud Connect copy (July 7 restore point)	PASS	The off-site copy is restorable over the WAN. Two independently verified copies of KOR-FS01's pre-incident state exist.
3. KOR-FS01 file-level restore from the new Synology backup (post-deduplication, July 20)	FAIL	Cause identified and confirmed — see below.

Scenario 3 — cause established.

"Restoring files from deduplicated volumes (E:\) requires that the mount server associated with the source backup repository uses Windows Server 2012 R2 or later with data deduplication feature enabled."

- The Data Deduplication feature was installed on KOR-BK01 (Windows Server 2019) and the server rebooted. **The error persisted**, which eliminates the simple remedy and confirms a format incompatibility: a Server 2019 mount server cannot read a chunk store created by **Windows Server 2025**.
- A **Windows Server 2025 mount server** is therefore required. KOR-APP01 (Server 2025, deduplication feature already installed) is the available candidate.
- Veeam 12.0 cannot add a Server 2025 machine as a managed server, so a **version upgrade is a prerequisite**.

Current recovery capability: file-level restore of KOR-FS01 is available for data dated **before 2026-07-13** (from E: or the Cloud Connect copy). For data created after that date, recovery is via Instant VM Recovery from the Synology backup. Full-VM and volume restores are unaffected throughout.

5. Repository capacity analysis

The KOR-FS01 chain on the Default repository was examined against both the Veeam configuration database and the files on disk:

File	Size	Role
Kor-FS01....2026-07-15...vbk	14.7 TB	Chain head (current full) — required by every restore point in the chain
Kor-FS01....2026-07-07...vrb	8.3 TB	July 7 restore point, enlarged by the deduplication rewrite — required
17 × .vrb (Jun 20 – Jul 7)	~87 GB	Daily restore points — required
9 × .vrb dated Dec 2025	~44 GB	Not referenced by any restore point — reclaimable

- KOR-FS01's chain occupies approximately **23 TB of the 25.4 TB repository**. In reverse-incremental mode the .vbk holds the current state and all earlier points are reconstructed by rolling backward from it, so **no material capacity can be released without removing the entire local KOR-FS01 history**, including the pre-deduplication July 1–7 points. Only ~44 GB is safely reclaimable today.
- The 2026-07-15 "Full" restore point originates from the interrupted session and its integrity is **unverified**. We suggest validating it (Veeam Backup Validator) before any decision on the chain.
- Capacity headroom, for context: a 21.1 TB source in a reverse-incremental chain requires the repository to hold a full plus ongoing rollback growth. We would value T-Net's view on target sizing (question 4, §6).

6. Open items — guidance requested

We would value T-Net's recommendation on the following. Our aim is to agree a design that is durable rather than to revisit individual events.

1. **Job design.** Given the measured difference between the two approaches (§3), what backup mode and target would you recommend for KOR-FS01 and for the remaining VMs going forward? Should Kor-VMs-New also move to forward incremental?
2. **Version upgrade and licensing.** We plan to upgrade from 12.0.0.1420. **Does our rental licence entitle v13 (current: 13.0.2.29), or should we target the latest 12.3.x?** Is a direct upgrade from 12.0 supported, or is an intermediate step required? Are you able to supply the ISO, or would you prefer to perform the upgrade?
3. **Mount server.** We propose adding KOR-APP01 (Server 2025) as a managed server and assigning it as mount server for the Synology repository, to restore file-level recovery for KOR-FS01. Do you concur, or would you prefer a dedicated mount server?
4. **Repository capacity.** What capacity and configuration would you recommend for the Default repository (or its replacement), given a 21 TB file server that continues to grow? Is there a refresh plan we should be aware of?
5. **The existing E: chain (~23 TB).** Would you recommend validating and retaining it, or retiring it once the Synology and Cloud Connect copies are established? We would prefer not to release the pre-deduplication history until a path is agreed.
6. **Off-site coverage for KOR-FS01.** Now that KOR-FS01 has a dedicated chain, it no longer flows to the Cloud Connect copy (its July 7 restore points remain). Approximately 5.8 TB would need to reach the DC. What is your recommended approach and timeline for the seed, and who would perform it?
7. **Monitoring and alerting.** We have configured Veeam email notifications to KOR. We would like to align on which alerts your monitoring raises, the expected response, and how job failures should reach us — so that a repeat of an extended undetected failure is not possible.
8. **Certificate lifecycle.** The vCenter certificate now expires 2028-07-14. Could we agree ownership and a reminder mechanism ahead of that date?

9. **ManualNew job.** Enabled but unscheduled and never run; targets Synology105 and contains a March 2026 KOR-FS01 full. We would like to confirm its origin and whether it should be retained, repurposed or retired.
10. **Restore testing.** We would like to establish a regular schedule (proposed: quarterly file-level, annual full-VM). Is this something you would run, or should KOR?
11. **Environment tidy-up.** The following are pending your input: decommissioned BMZ-HO-APP002 / DC005 / FS007 / RDS001 folders on UC3200 (FS007 is the former file server — we will not delete without confirming no unmigrated data); duplicate Kor-RDS01 / Kor-RDS01_1 folders and the corresponding duplicate backup entries; and orphaned catalogue records (August2024 , Kor-VMs).

7. Proposed next steps

#	Step	Owner	Timing
1	Confirm licence entitlement and supported upgrade path; supply or approve ISO	T-Net	On receipt
2	Upgrade Veeam (configuration backup and VM snapshot taken beforehand)	KOR or T-Net	Agreed window
3	Add KOR-APP01 as managed server; assign as mount server for Synology105	KOR	After step 2
4	Re-test file-level restore of the post-deduplication KOR-FS01 backup	KOR	After step 3
5	Validate the 2026-07-15 restore point (Backup Validator)	KOR / T-Net	After step 2
6	Agree job design and repository sizing; implement	T-Net / KOR	Following review
7	Establish off-site coverage for the KOR-FS01 chain, including seed	T-Net / KOR	Scheduled
8	Agree monitoring, alerting and restore-test cadence	T-Net / KOR	Following review
9	Decide the disposition of the existing E: chain	T-Net / KOR	After steps 4 and 5
10	Environment tidy-up (decommissioned VMs, duplicate folders, orphaned records)	KOR with T-Net input	Following review

Prepared by KOR Structural Operations, 2026-07-20. All figures verified from the Veeam console, Veeam configuration database, repository filesystems and vSphere between 2026-07-15 and 2026-07-20. Configuration changes are limited to those listed in §3. We are happy to walk through any of the underlying evidence — session logs, restore-point listings and error output — on request.